Exercises

Exercise 9.1 · Computations. Let $A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 4 & -2 \end{pmatrix}$. Compute $A + B$, $2A - B$, AB and BA.

Solution. $A+B = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$, $2A-B = \begin{pmatrix} 2 & 3 \\ -6 & 8 \end{pmatrix}$, $AB = \begin{pmatrix} 8 & -3 \\ 12 & -7 \end{pmatrix}$, $BA = \begin{pmatrix} -1 & 3 \\ 6 & 2 \end{pmatrix}$. We see that $AB \neq BA$.

Exercise 9.2 · Inverse and system of order 2. Let $A = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$.

1. Explain why A is invertible and find A^{-1}.
2. Deduce the solution of the system $3x + 5y = 4$, $x + 2y = 1$.

Solution.

1. $\det A = 6 - 5 = 1 \neq 0$ and $A^{-1} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}$.
2. $X = A^{-1}\begin{pmatrix} 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 8-5 \\ -4+3 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$: $x = 3$, $y = -1$.

Exercise 9.3 · System of order 3. Consider the system $x+y+z = 6$, $2x-y+z = 3$, $x+2y-z = 2$.

1. Write it in the form $AX = B$.
2. With a calculator, find A^{-1}, then the solution of the system.
3. Check the solution by hand.

Solution.

1. The system can be written $AX = B$ with

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{pmatrix}, \qquad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \qquad B = \begin{pmatrix} 6 \\ 3 \\ 2 \end{pmatrix}.$$

2. The calculator gives

$$A^{-1} = \frac{1}{7}\begin{pmatrix} -1 & 3 & 2 \\ 3 & -2 & 1 \\ 5 & -1 & -3 \end{pmatrix} \qquad \text{and} \qquad X = A^{-1}B = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

3. $1 + 2 + 3 = 6$, $2 - 2 + 3 = 3$, $1 + 4 - 3 = 2$.

Exercise 9.4 · Powers by induction. Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. Compute A^2 and A^3, conjecture an expression of A^n for $n \geq 1$, then prove it by induction.

Solution. $A^2 = \begin{pmatrix} 4 & 4 \\ 0 & 4 \end{pmatrix}$, $A^3 = \begin{pmatrix} 8 & 12 \\ 0 & 8 \end{pmatrix}$. Conjecture: $A^n = \begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}$. True for $n = 1$. If it holds for n:

$$A^{n+1} = \begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2^{n+1} & 2^n + n2^n \\ 0 & 2^{n+1} \end{pmatrix},$$

and $2^n + n2^n = (n+1)2^n$: the formula holds for $n+1$.

Exercise 9.5 · A polynomial relation. Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

1. Check that $A^2 = 2A + 3I_2$.
2. Deduce that A is invertible and express A^{-1} in terms of A and I_2. Check with the formula of the course.

Solution.

1. $A^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ and $2A + 3I_2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$.
2. $A^2 - 2A = 3I_2$ can be written $A \times \frac{1}{3}(A - 2I_2) = I_2$, and likewise $\frac{1}{3}(A - 2I_2)A = I_2$. So $A^{-1} = \frac{1}{3}(A - 2I_2) = \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix}$. With the course: $\det A = -3$ and $A^{-1} = -\frac{1}{3}\begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}$, the same result.

Exercise 9.6 · Rotations.

1. Using the addition formulas, prove that $R_\theta R_{\theta'} = R_{\theta+\theta'}$.
2. Deduce $\left(R_{\pi/6}\right)^{12}$.
3. Find the image of the point $(2, 0)$ under the rotation with centre O and angle $\frac{\pi}{3}$.

Solution.

1. The entry in row 1 and column 1 of the product is $\cos\theta\cos\theta' - \sin\theta\sin\theta' = \cos(\theta + \theta')$; the entry in row 2 and column 1 is $\sin\theta\cos\theta' + \cos\theta\sin\theta' = \sin(\theta+\theta')$. The other two are obtained in the same way.
2. By induction, $\left(R_{\pi/6}\right)^{12} = R_{2\pi} = I_2$.
3. $R_{\pi/3}\begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 2\cos\frac{\pi}{3} \\ 2\sin\frac{\pi}{3} \end{pmatrix} = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}$: the point $(1, \sqrt{3})$.

Exercise 9.7 · A parabola through three points. Find the real numbers a, b, c such that the parabola with equation $y = ax^2 + bx + c$ passes through the points $(1, 2)$, $(2, 3)$ and $(3, 6)$. Write the system in matrix form.

Solution. The system is

$$\begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}.$$

Subtracting consecutive equations: $3a + b = 1$ and $5a + b = 3$, so $a = 1$, $b = -2$, then $c = 3$. Check: $1 - 2 + 3 = 2$, $4 - 4 + 3 = 3$, $9 - 6 + 3 = 6$.

Problem

Problem · The Fibonacci sequence and Cassini's identity. The Fibonacci sequence is defined by $F_0 = 0$, $F_1 = 1$ and $F_{n+2} = F_{n+1} + F_n$ for every $n \geq 0$. Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$.

1. Compute $F_2, \dots, F_6$ and A^2, A^3.
2. Prove by induction that, for every $n \geq 1$, $A^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$.
3. Prove that for all matrices M and N of order 2, $\det(MN) = \det M \times \det N$. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $N = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$.
4. Deduce that $\det(A^n) = (-1)^n$, then Cassini's identity: $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$ for $n \geq 1$.
5. Check the identity for $n = 5$.
6. Using $A^{2n} = A^n \times A^n$, prove that $F_{2n} = F_n(F_{n+1} + F_{n-1})$ for $n \geq 1$.

Beyond the syllabus · Determinant of a product. The relation $\det(MN) = \det M \times \det N$ of question 3 is not part of the syllabus. It is proved here by computation for matrices of order 2 and is not assessed.

Solution · Problem.

1. $F_2 = 1, F_3 = 2, F_4 = 3, F_5 = 5, F_6 = 8$. $A^2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $A^3 = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$.
2. For $n = 1$: $\begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = A$. If the formula holds for n, then
$$A^{n+1} = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} F_{n+1}+F_n & F_{n+1} \\ F_n + F_{n-1} & F_n \end{pmatrix} = \begin{pmatrix} F_{n+2} & F_{n+1} \\ F_{n+1} & F_n \end{pmatrix}.$$
3. $MN = \begin{pmatrix} aa'+bc' & ab'+bd' \\ ca'+dc' & cb'+dd' \end{pmatrix}$. Its determinant is $(aa'+bc')(cb'+dd') - (ab'+bd')(ca'+dc')$. After expanding, the terms $aa'cb'$ and $bc'dd'$ cancel with their opposites, and what remains is $ada'd' + bcb'c' - adb'c' - bca'd' = (ad-bc)(a'd'-b'c')$.
4. By induction, $\det(A^n) = (\det A)^n = (-1)^n$. With question 2, $\det(A^n) = F_{n+1}F_{n-1} - F_n^2$.
5. $F_6F_4 - F_5^2 = 24 - 25 = -1 = (-1)^5$.
6. The entry in row 1 and column 2 of A^{2n} is F_{2n}; that of A^nA^n is $F_{n+1}F_n + F_nF_{n-1} = F_n(F_{n+1} + F_{n-1})$.

Teacher note · Status of the problem. The determinant of order 2 is used for the inverse; its multiplicativity is not in the syllabus and is proved here by computation. Possible extension: the Hill cipher, which encrypts pairs of letters with a matrix of order 2 invertible modulo 26, listed among the possible problems of the syllabus.